MMBench: Is Your Multi-modal Model an All-around Player?

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01 Why Do We Need MMBench?

Limitations Of Existing Benchmarks

Traditional VQA/caption datasets suffer false negatives and weak instruction-following handling

Lack fine-grained ability diagnostics across perception and reasoning

Human-in-the-loop evaluations are biased, costly, irreproducible, and unscalable

Gaps hinder rigorous, holistic, comparable assessment of modern VLMs

Goals Of MMBench

Provide objective, scalable, bilingual benchmark with rigorous metrics

Enable fine-grained, ability-centric diagnostics across perception and reasoning

Reduce bias/instability via quality control and robust protocol

Support weak instruction-following via LLM-based choice extraction

Enable apples-to-apples English–Chinese comparisons

02 Related Work And Gaps

Multimodal Datasets Landscape

Existing datasets target selective tasks: captioning, VQA, OCR, science, video

Task-specific formats/metrics hinder cross-ability evaluation and comparability

Broader efforts (OwlEval, MME) remain limited in scale or objectivity

Multimodal Models Overview

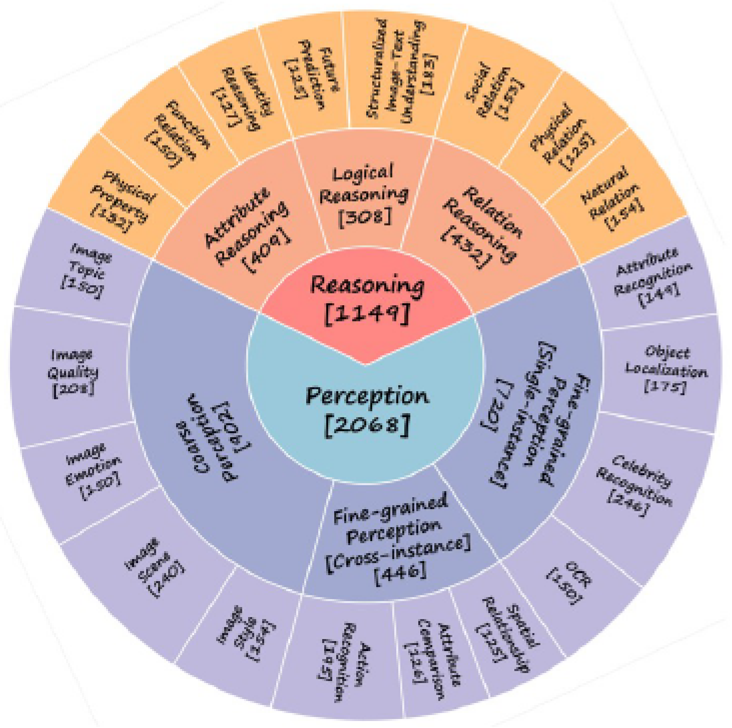
Models span Flamingo, BLIP-2, LLaVA, mPLUG-Owl, CogVLM, Qwen-VL, Yi-VL, etc.

Proprietary GPT-4v, Gemini-Pro-V, Qwen-VL-Max included

Instruction tuning and stronger LLM backbones drive performance

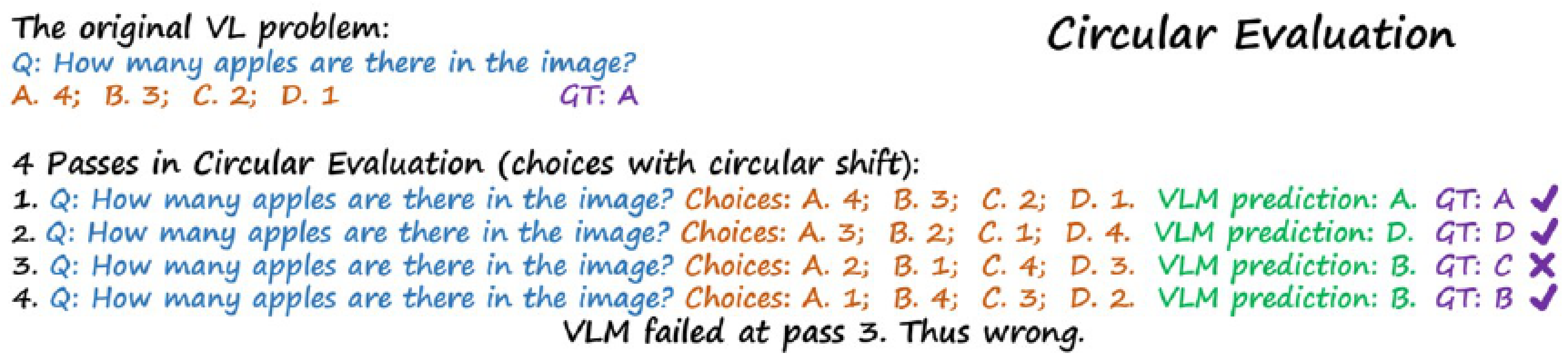
Benchmark needed to expose ability-level strengths and weaknesses

03 Systematically Constructed Dataset



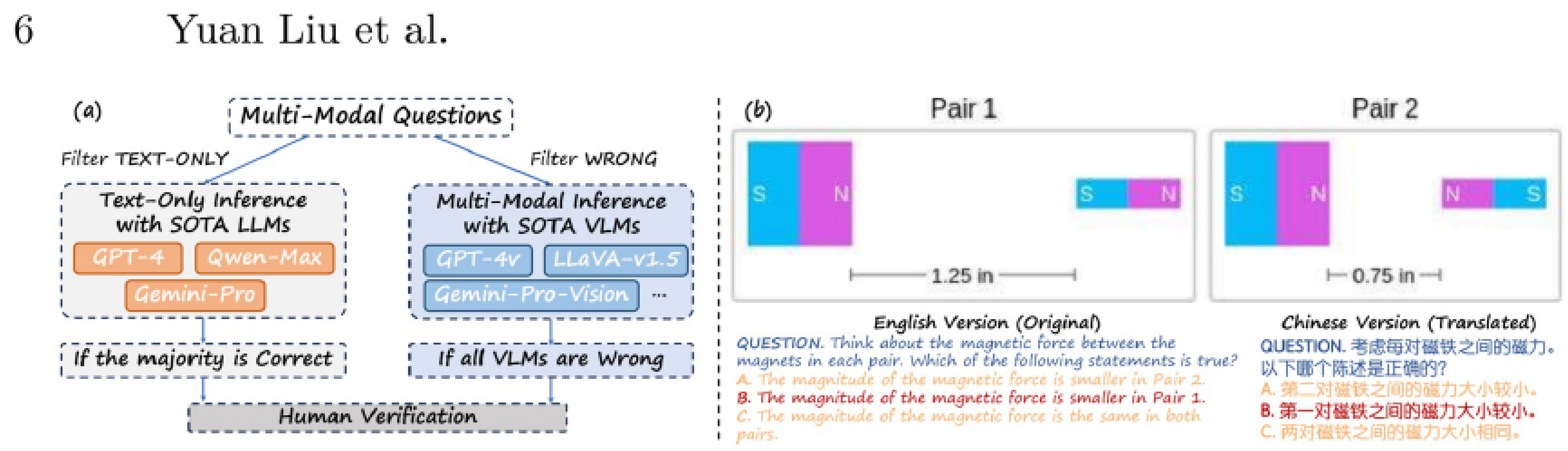
- 3,217 curated MCQs spanning 20 fine-grained abilities
- Hierarchical taxonomy across Perception and Reasoning
- Balanced samples per leaf ability (≈125+) enable robust diagnostics
- Focus on ability-level insights beyond task aggregates

03 Robust, Scalable Evaluation



- CircularEval reduces guess bias via multi-pass consistency
- LLM-based choice extraction standardizes free-form outputs
- Protocol exposes stability gaps and improves comparability

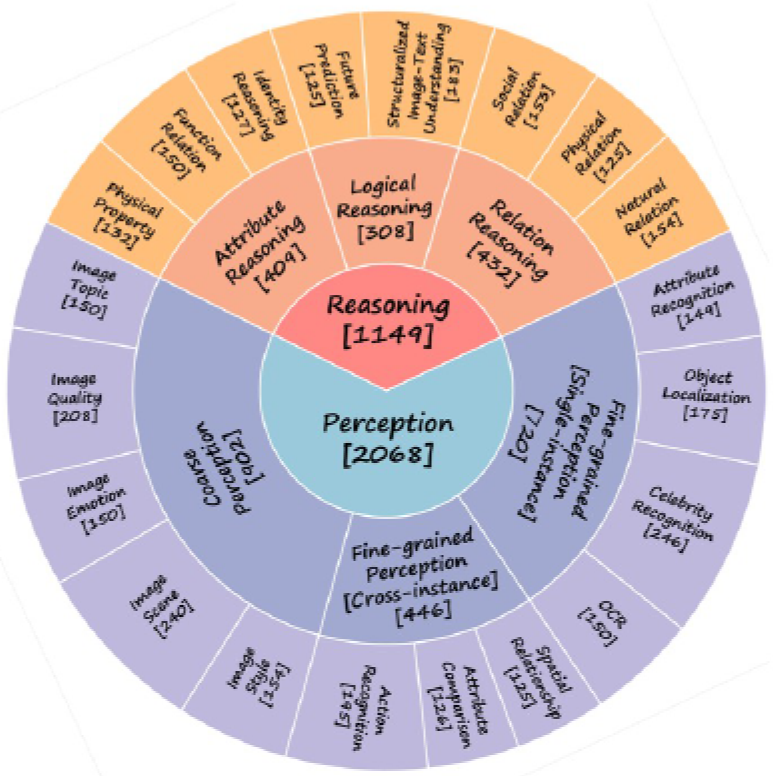
03 Bilingual Benchmarking



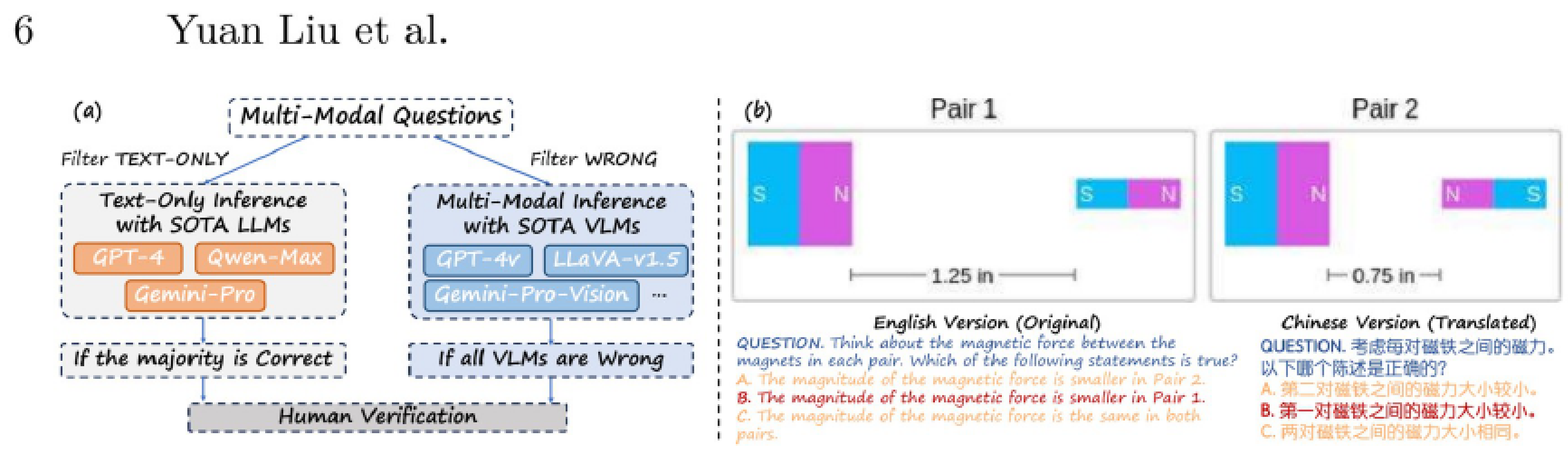
- MMBench-CN translates while preserving proper nouns/symbols
- Human-verified translations enable direct EN–CN comparisons
- Probes language balance in training and tuning

04 Hierarchical Ability Taxonomy

- Level-1: Perception and Reasoning
- Six level-2 groups, 20 level-3 leaf abilities
- Covers attributes, logic, relations, single/cross-instance perception
- Mirrors human-like multimodal competencies



04 Data Collection Process



- MCQs as tuples: question, choices, image, answer
- 80% sourced online; remainder paired images with constructed questions
- Seed sets guided balanced, diverse expansion

04 Quality Control Strategies

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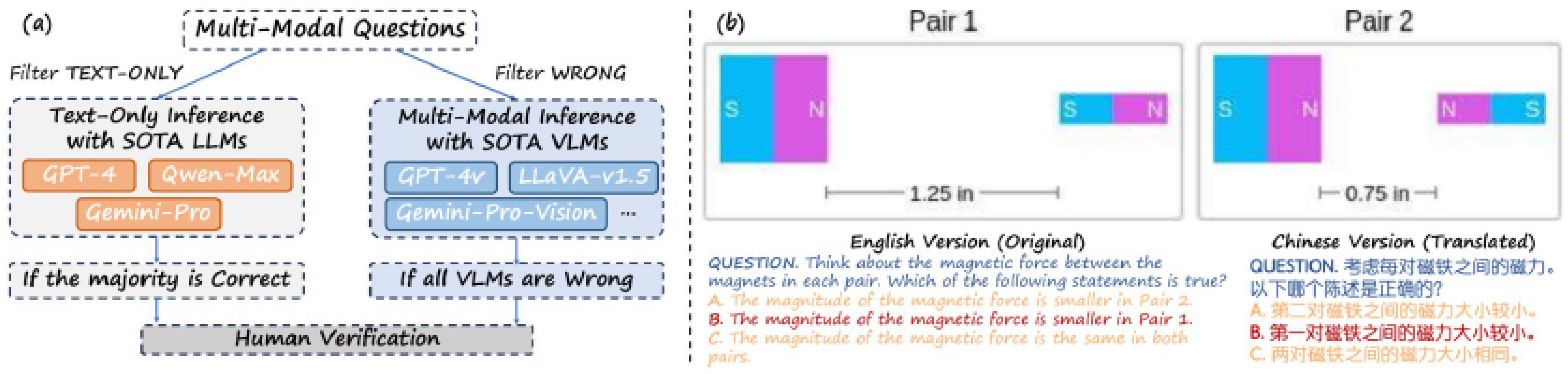
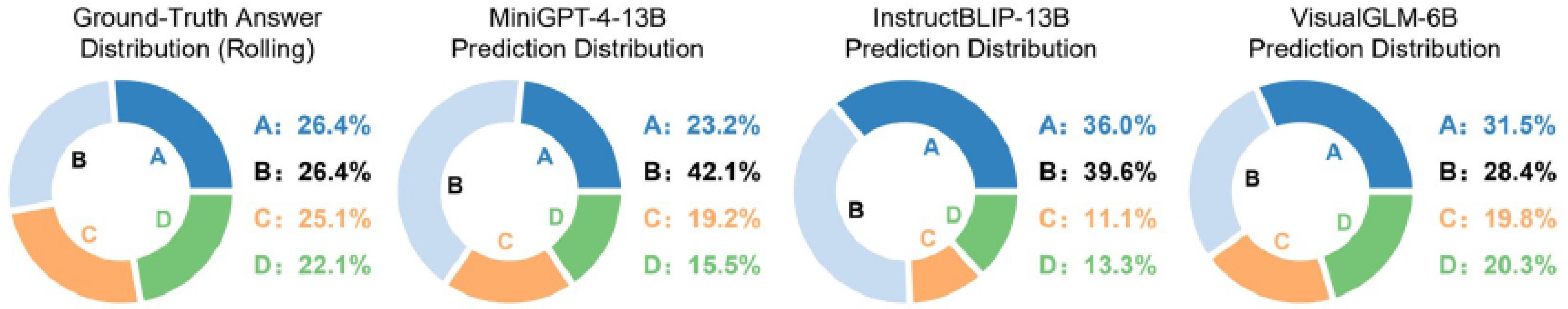


- Text-only solvability detection via top-LLM majority vote
- Wrong-sample detection: items all strong VLMs miss
- Manual review removes flawed or text-solvable items

04 Data Splits And Access

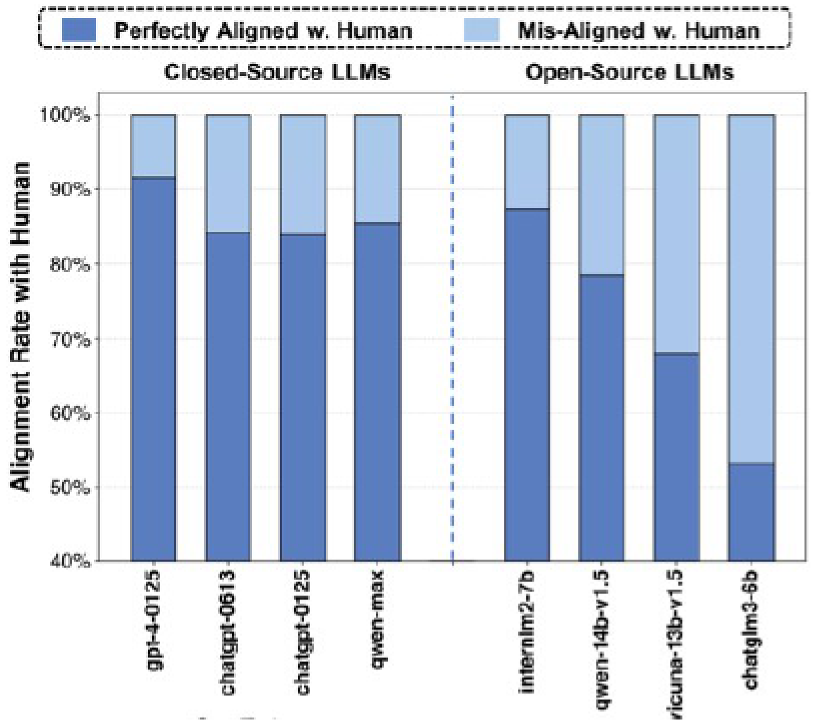
- 4:6 dev:test split
- Dev labels public; test labels server-held for blind eval
- Prevents overfitting while enabling reproducible comparisons

05 Step 1: Heuristic Choice Matching



- Parse outputs for A–D labels for compliant models
- Direct comparison enables fast standardized scoring
- Serves as quick first-pass evaluation

05 Step 2: LLM-Based Choice Extraction



- Use LLM to align free-form predictions to closest option
- Outputs label or sentinel Z for no-match
- GPT-4 achieves 91.5% human alignment

05 CircularEval: Robust Multi-Pass Testing

Circular Evaluation

The original VL problem:
Q: How many apples are there in the image?
A. 4; B. 3; C. 2; D. 1 GT: A

4 Passes in Circular Evaluation (choices with circular shift):
1. Q: How many apples are there in the image? Choices: A. 4; B. 3; C. 2; D. 1. VLM prediction: A. GT: A ✓
2. Q: How many apples are there in the image? Choices: A. 3; B. 2; C. 1; D. 4. VLM prediction: D. GT: D ✓
3. Q: How many apples are there in the image? Choices: A. 2; B. 1; C. 4; D. 3. VLM prediction: B. GT: C ✗
4. Q: How many apples are there in the image? Choices: A. 1; B. 4; C. 3; D. 2. VLM prediction: B. GT: B ✓

VLM failed at pass 3. Thus wrong.

- Test N times with circularly shifted choices/answers
- Success requires correctness on all passes
- Early stopping balances robustness and cost

06 Models And Inference Settings

- Evaluate wide range: open-source VLMs and proprietary GPT-4v, Gemini, Qwen-VL
- Zero-shot, uniform prompts, open-ended generation
- gpt-4-0125 used for choice extraction
- Ensures fairness across models

06 Instruction-Following Variability

Model Name	Match Rate	DEV Acc	Model Name	Match Rate	DEV Acc
MiniGPT4-7B	85.7	47.9 +8.8	MiniGPT4-13B	84.8	52.1 +8.7
InstructBLIP-7B	93.6	57.1 +4.3	InstuctBLIP-13B	93.7	58.4 +5.6
IDEFICS-9B-Instruct	96.6	58.4 +1.5	Qwen-VL-Chat	93.8	73.3 +3.6
MiniCPM-V	95.2	70.9 +4.5	VisualGLM-6B	64.8	39.9 +23.2
GPT-4v	91.8	81.5 +3.6	GeminiProVision	97.5	81.8 +0.8
Qwen-VL-Plus	77.4	64.5 +15.0	Qwen-VL-Max	96.0	82.0 +3.2

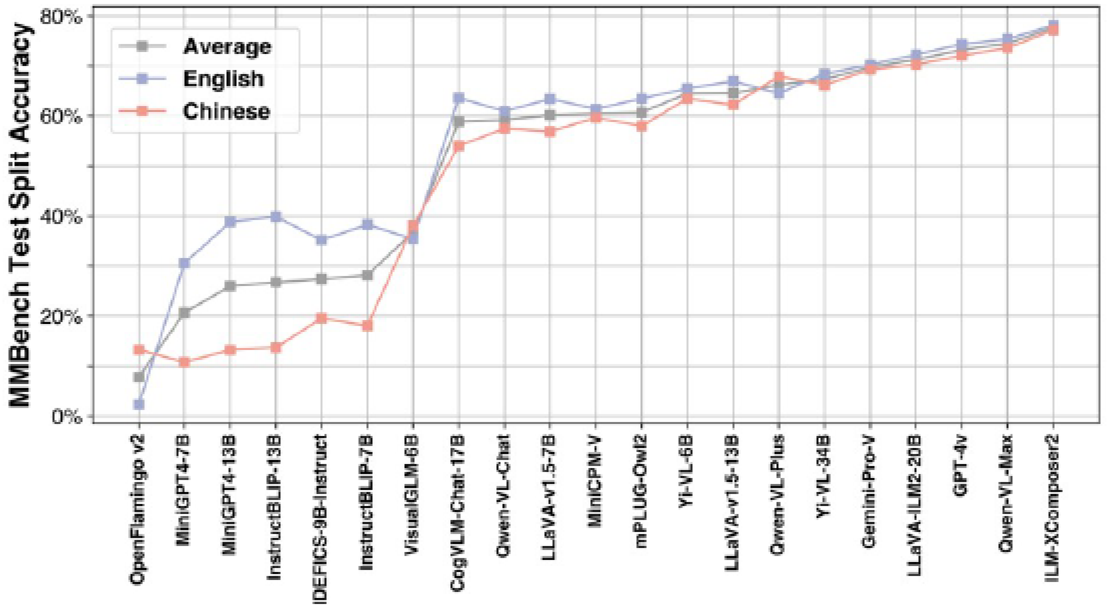
- Heuristic match rates vary widely across models
- LLM extraction boosts accuracy for low-compliance models
- Decouples instruction-following from multimodal competence

07 CircularEval Versus Single-Pass

VLM	Circular	Acc Change	VLM	Circular	Acc Change	VLM	Circular	Acc Change
MiniGPT4-7B	32.7%	-24.1%	MiniGPT4-13B	37.5%	-23.2%	Yi-VL-6B	65.6%	-9.8%
InstructBLIP-7B	37.4%	-24.0%	InstructBLIP-13B	40.9%	-23.0%	Yi-VL-34B	68.2%	-9.5%
LLaVA-v1.5-7B	62.5%	-11.2%	LLaVA-v1.5-13B	67.2%	-8.6%	MiniCPM-V	64.8%	-10.6%
IDEFICS-9B-Instruct	37.2%	-22.6%	LLaVA-InternLM2-20B	72.8%	-7.0%	Qwen-VL-Plus	62.9%	-16.6%
VisualGLM-6B	36.1%	-27.0%	CogVLM-Chat-17B	62.4%	-15.6%	Qwen-VL-Max	76.4%	-8.7%
Qwen-VL-Chat	59.5%	-17.4%	mPLUG-Owl2	63.5%	-8.7%	Gemini-Pro-V	70.9%	-11.7%
OpenFlamingo v2	2.6%	-34.1%	InternLM-XComposer2	79.1%	-4.7%	GPT-4v	74.3%	-10.8%

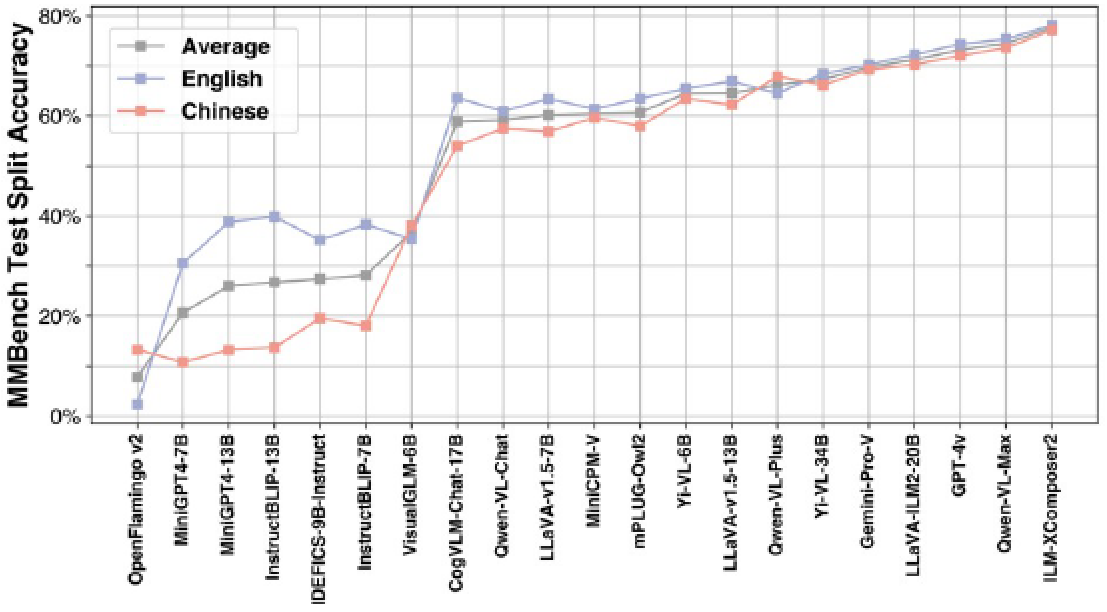
- CircularEval lowers accuracies, widening performance gaps
- Drops of 8–27% common, even for top models
- Reveals sensitivity to option bias and perturbations

07 Overall Performance Highlights



- InternLM-XComposer2 leads open-source; LLaVA and Yi-VL strong
- Proprietary GPT-4v and Qwen-VL-Max competitive
- Small models like MiniCPM-V notable; OpenFlamingo v2 lags

07 Bilingual Evaluation Findings



- Most models score lower on MMBench-CN vs English
- Top performers show small EN–CN gaps
- Balanced bilingual LLMs reduce drops

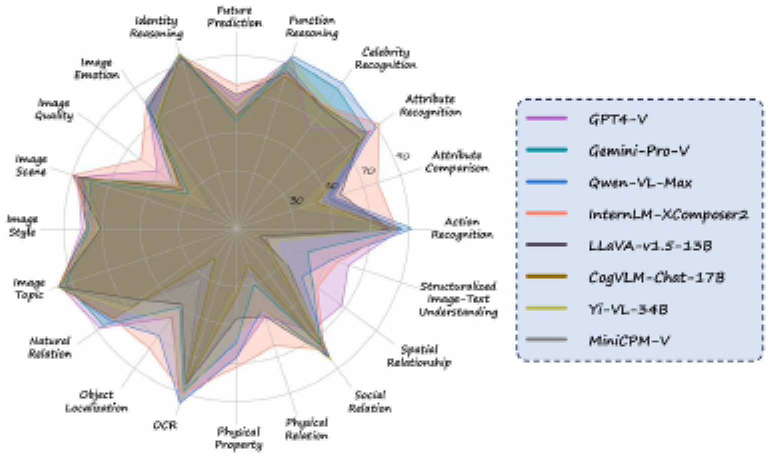
07 Content Moderation Effects

- Proprietary systems sometimes refuse answers, slightly lowering scores
- Upper-bound recalculations show ≤2.4% absolute impact
- Refusal patterns vary by provider and category

Model	MMBench-test	Upper Bound
GPT-4v	74.3	76.2
Gemini-Pro-V	70.2	72.6
Qwen-VL-Max	75.4	75.5

08 Ability-Wise Strengths And Weaknesses

Model	Overall	CP	FP-S	FP-C	AR	LR	RR
Large Language Models							
GPT-4-Turbo (0125) [26]	2.9%	0.6%	1.2%	4.1%	3.7%	4.9%	7.4%
OpenSource VLMs							
OpenFlamingo v2 [3]	2.3%	1.1%	3.5%	1.5%	5.3%	0.0%	2.7%
MiniGPT4-7B [41]	30.5%	37.0%	31.8%	17.2%	49.8%	9.2%	25.6%
IDEFICS-9B-Instruct [19]	35.2%	48.3%	31.3%	29.6%	47.8%	11.4%	25.2%
VisualGLM-6B [11]	35.4%	40.2%	38.5%	26.2%	47.8%	19.6%	29.5%
InstructBLIP-7B [9]	38.3%	46.7%	39.0%	31.8%	55.5%	8.7%	31.0%
MiniGPT4-13B [41]	38.8%	44.6%	42.9%	23.2%	64.9%	8.2%	32.9%
InstructBLIP-13B [9]	39.8%	47.2%	42.9%	21.0%	60.4%	12.5%	38.8%
Qwen-VL-Chat* [4]	60.9%	68.5%	67.7%	50.2%	78.0%	37.0%	45.7%
MiniCPM-V [27]	61.4%	65.6%	69.4%	51.3%	70.6%	35.3%	59.7%
LLaVA-v1.5-7B [21]	63.4%	70.0%	68.0%	57.7%	77.6%	33.2%	56.2%
mPLUG-Owl2 [37]	63.5%	68.1%	69.1%	55.8%	78.4%	37.0%	57.0%
CogVLM-Chat-17B [34]	63.6%	72.8%	66.6%	53.4%	71.4%	33.7%	62.0%
Yi-VL-6B* [1]	65.5%	72.8%	72.9%	56.2%	73.5%	41.3%	55.4%
LLaVA-v1.5-13B [21]	66.9%	73.1%	72.4%	60.3%	73.5%	35.9%	65.5%
Yi-VL-34B* [1]	68.4%	72.0%	78.0%	54.7%	81.2%	38.6%	68.2%
LLaVA-InternLM2-20B [8]	72.3%	78.3%	76.6%	68.2%	78.4%	46.2%	69.4%
InternLM-XComposer2* [10]	78.1%	80.4%	83.5%	73.0%	83.7%	63.6%	74.4%
Proprietary VLMs							
Qwen-VL-Plus [4]	64.6%	66.5%	79.1%	50.2%	73.9%	42.9%	57.8%
Gemini-Pro-V [31]	70.2%	70.0%	78.9%	65.9%	82.9%	46.2%	65.9%
GPT-4v [26]	74.3%	77.6%	73.8%	71.5%	85.3%	63.6%	68.6%
Qwen-VL-Max [4]	75.4%	74.8%	87.2%	67.0%	85.3%	54.9%	70.5%



- Proprietary models excel in structured and knowledge-heavy tasks
- Open-source leaders match/exceed on many abilities
- LLM choice/scale boost logic and relation reasoning

08 Common Failure Modes

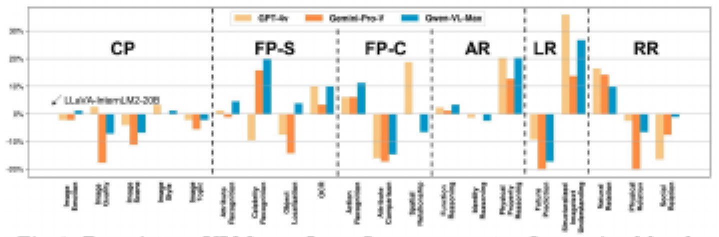
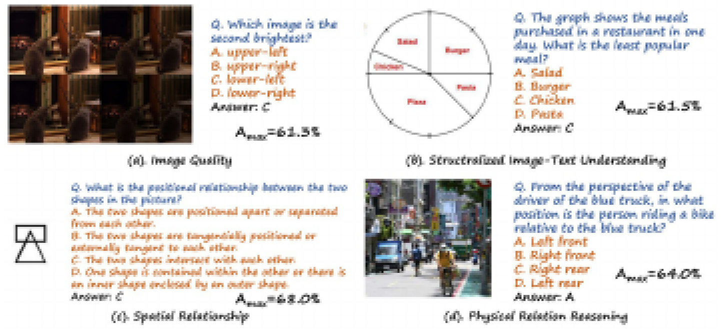


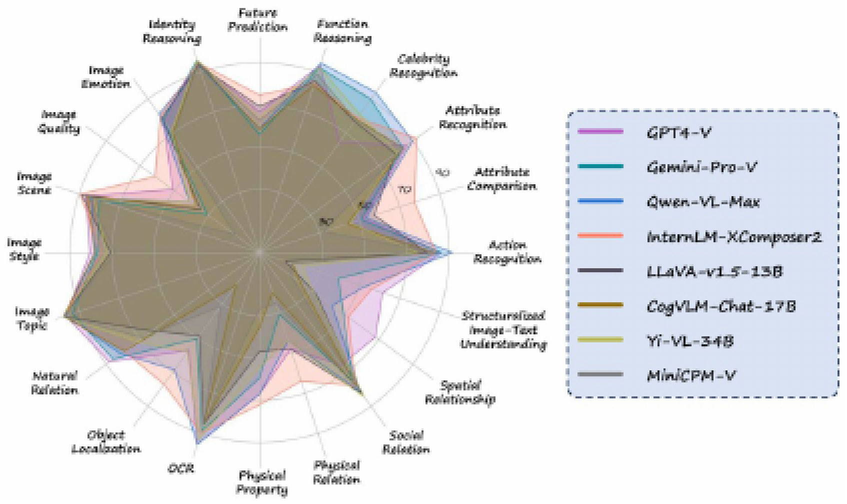
Fig. 9: Proprietary VLMs vs. Open-Source ones at a fine-grained level.



- Difficult: low-level visual comparisons (brightness, sharpness, artifacts)
- Challenges with structured inputs: tables, diagrams, layouts
- Weak precise 2D/3D spatial relation reasoning

09 MMBench Impact And Next Steps

- Delivers scalable, objective, bilingual benchmark with fine-grained abilities
- CircularEval and LLM extraction reveal clear capability profiles
- Future: expand abilities, enrich structured content, refine bilingual balance
- Explore stricter robustness and automated data quality pipelines



THANKS!